QP Code: **Reg No……..**

Name ……...

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP BPES (HONOURS)

FIFTH SEMESTER EXAMINATION

MODEL QUESTION PAPER

COURSE NAME: PHYSIOLOGY OF EXERCISE

Course Code: MG5DSCPES300

Duration: 2 Hours **Maximum Marks: 70**

Part A

Multiple Choice Questions.

Answer All Questions

Each question carries 1 mark (10 × 1 = 10)

1. The immediate energy system, which provides ATP for about 10-15 seconds of maximal effort is the [K, CO2]

a) Oxidative system b) Glycolytic system
c) ATP-PC system d) Beta-oxidation system

2. The process of breaking down triglycerides into fatty acids and glycerol for energy is called ……. [K, CO5]

a) Glycogenesis b) Lipolysis c) Gluconeogenesis d) Ketogenesis

3. At the neuromuscular junction, the neurotransmitter released from the motor neuron to stimulate skeletal muscle contraction is …….. [K, CO 3]

a) Norepinephrine b) Dopamine
c) Acetylcholine d) Serotonin

4. The "second wind" phenomenon during prolonged exercise is best described as……. [U, CO 3]

a) A sudden drop in core body temperature
b) A subjective feeling of relief and reduced dyspnea after initial discomfort
c) A massive spike in blood lactate concentration
d) The complete depletion of muscle glycogen stores

5. Essential fat is necessary for normal physiological function and is found in ……. [K, CO 6]

a) Subcutaneous tissue only b) Visceral depots only
c) Bone marrow, heart, lungs, and CNS d) Adipose tissue exclusively

6. Body composition refers to the relative proportions of ……. [K, CO 6]

a) Water and protein only b) Fat mass and fat-free mass
c) Bone density and muscle mass only d) Intracellular and extracellular fluid

7. The gender difference in hemoglobin concentration typically results in females having a lower [U, CO 7]

a) Resting heart rate b) Arteriovenous oxygen difference
c) Total peripheral resistance d) Blood glucose level

8. Which of the following cardiovascular variables is typically higher at rest in females compared to males? [U, CO 7]

a) Stroke volume b) Cardiac output
c) Resting heart rate d) Blood volume

9. The age-related loss of skeletal muscle mass and strength is termed as ……. [K, CO 1,2]

a) Osteopenia b) Sarcopenia
c) Dynapenia d) Atrophy

10. The "talk test" is a practical tool for exercising pregnant women and older adults to ……[U, CO 1,6]
    a) Assess respiratory muscle strength b) Monitor exercise intensity
    c) Measure lung capacity d) Evaluate verbal cognitive function

Part B

Short Answer Type Questions

Answer any 5 Questions

Each question carries 2 marks (5 × 2 = 10)

11. Define excess post-exercise oxygen consumption. [K, CO 2]
12. List four specific physiological adaptations that occur in skeletal muscle following a chronic endurance training program. [K, CO 3]
13. Explain the physiological mechanism behind the "diaphragmatic fatigue" that can occur during very prolonged endurance events? [U, CO 3]
14. Distinguish between essential fat and storage fat. [U, CO 6]
15. Describe the differences between males and females in terms of body fat distribution. [U, CO 7]
16. What is the FITT-VP principle for prescribing exercise to older adults? [U, CO 1,2]
17. What is "hypoglycemia unawareness" in diabetic patients? [U, CO 2,6]

Part C

Short Essay Type Questions

Answer any 4 Questions

Each question carries 5 marks (4 × 5 = 20)

18. Discuss how all three energy systems contribute to ATP production during a 400m sprint. [An, CO 2,5]
19. Discuss how anaerobic training affects the nervous system. [U, CO 3]
20. Describe the rationale and standard procedure for using skinfold calipers to estimate body composition. [Ap, CO 6]
21. Explain the physiological differences between males and females in thermoregulation during exercise in a hot environment. [An, CO 3]
22. Discuss the physiological benefits of regular physical activity during a normal pregnancy for both the mother and the fetus. [U, CO 2,3]
23. Explain the differences in endurance training adaptations between males and females. [An, CO 7]

Part D

Essay Type Questions

Answer any 2 Questions

Each question carries 15 marks (2 × 15 = 30)

24. Explain how pulmonary ventilation, diffusion gradient, and blood oxygen saturation are affected in hot, humid and high altitude conditions? [An, CO 4]
25. Design a practical protocol for monitoring body composition changes over a 6-month competition season. Include recommendations for standardization of measurement conditions. [C, CO 6]

26. Based on sex differences in substrate metabolism, recommend distinct carbohydrate and fat fueling strategies for athletes during the marathon race. [C, CO 7]

27. Explain the different types of skeletal muscle fibers and discuss their structural and functional characteristics. How do these fiber types influence athletic performance? [An, CO 3]